

Kuwait University

2021/2022

Spring course



Calculus B

First

Sections : **7.4 (Part c/c) – 7.5 – 7.8**

7.4 Integration of Rational Functions by Partial Fractions.

7.5 Strategy for Integration.

7.8 Improper Integrals.

(7)



Calculus B Quiz #6



$$\int (e^{3x} + 2^x - e^{\frac{x}{2}} + 3^{6x}) dx$$

$$\int \sec^2 x e^{\tan x} dx$$

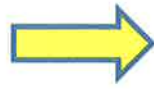
$$\int \frac{\cos x}{\sin^2 x + 4} dx$$

$$\int \frac{e^{\frac{1}{x}}}{x^3} dx$$





طريقة التفكير بالتكامل في الامتحان



التبسيط قبل الحل



هناك دالة بالمقام مشتقتها
موجود بالبسط

جذور

زاوية أو أس ليس حدودية
من الدرجة الأولى



ثم



By parts

عند وجود أولوية
من الأولويات

بس دوال مثلثية
أو (h)

حدودية من
الدرجة الثانية

u

dv

دوال مثلثية بالمقام
وما تصعد بالبسط

حدودية من
الدرجة الثانية $\frac{1}{2} \frac{3}{2} \frac{5}{2} \dots$

حدودية
حدودية

المقام < البسط

البسط ≤ المقام

Partial fraction

قسمة مطولة

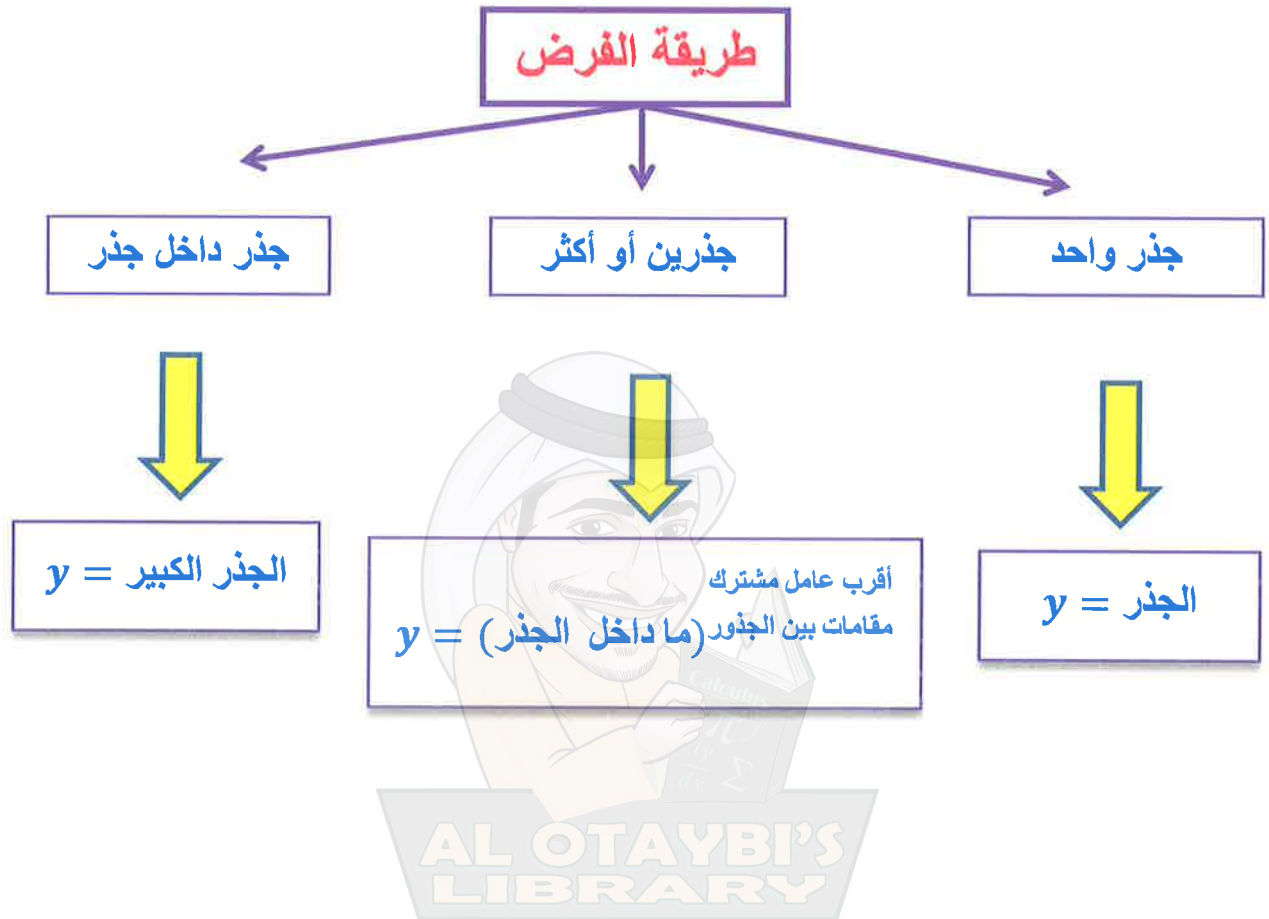


Miscellaneous (Root) substitution



عند وجود جذور (مشاكل بالأسس) , فانه يتم تبسيط السؤال قبل حله .

يجب أن يتم تنظيف السؤال قبل حله





Text book question



$$\int \frac{dx}{x^2 + x\sqrt{x}}$$

Let $y = \sqrt{x}$
 $y^2 = x$
 $2y dy = dx$

$$\int \frac{2y}{(y^2)^2 + y^2 y} dy = \int \frac{2y}{y^4 + y^3} dy$$

$$\int \frac{2y}{y^3 (y+1)} dy = \int \frac{2}{y^2 (y+1)} dy$$

$$\frac{2}{y^2 (y+1)} = \frac{A}{y} + \frac{B}{y^2} + \frac{C}{y+1}$$

* $y^2 (y+1)$

$$2 = A y (y+1) + B (y+1) + C y^2$$

$y=0$ $2 = 0 + B(1) + 0 \rightarrow B = 2$

$y=-1$ $2 = 0 + 0 + C(1) \rightarrow C = 2$

Equate the coeff of y^2

$$0 = A + C$$

$$0 = A + 2 \rightarrow A = -2$$

$$\int \left[\frac{-2}{y} + \frac{2}{y^2} + \frac{2}{y+1} \right] dy$$

$$= -2 \ln|y| + 2 \frac{y^{-1}}{-1} + 2 \ln|y+1| + C$$

$$= -2 \ln|\sqrt{x}| + 2 \frac{(\sqrt{x})^{-1}}{-1} + 2 \ln|\sqrt{x}+1| + C$$

$$\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$$

$$\int \frac{dx}{x^{\frac{1}{2}} + x^{\frac{1}{3}}}$$

$$\frac{\frac{1}{2}}{\frac{1}{6}}$$

Let

$$y = x^{\frac{1}{6}}$$

$$y^3 = x^{\frac{1}{6} \cdot 3} = x^{\frac{1}{2}}$$

$$y^2 = x^{\frac{1}{6} \cdot 2} = x^{\frac{1}{3}}$$

$$y^6 = x$$

$$6y^5 dy = dx$$

$$\int \frac{6y^5}{y^3 + y^2} dy = 6 \int \frac{y^5}{y^2(y+1)} dy$$

$$6 \int \frac{y^3}{y+1} dy$$

$$6 \int [y^2 - y + 1 + \frac{-1}{y+1}] dy$$

$$6 \left[\frac{y^3}{3} - \frac{y^2}{2} + y - \ln|y+1| \right] + C$$

$$6 \left[\frac{(x^{\frac{1}{6}})^3}{3} - \frac{(x^{\frac{1}{6}})^2}{2} + x^{\frac{1}{6}} - \ln|x^{\frac{1}{6}}+1| \right] + C$$

$$\begin{array}{r} y^2 - y + 1 \\ 1+y \overline{) y^3} \\ \underline{\ominus y^3 \oplus y^2} \\ -y^2 \\ \underline{\oplus -y^2 \oplus y} \\ -y \\ \underline{\ominus -y \oplus 1} \\ -1 \end{array}$$

$$\int \frac{\sqrt[4]{x+9}}{1+\sqrt{x+9}} dx$$

$$\int \frac{(x+9)^{\frac{1}{4}}}{1+(x+9)^{\frac{1}{2}}} dx$$

$$\frac{1}{4} \quad \frac{1}{2}$$

$$\left(\frac{1}{4}\right)$$

Let $y = (x+9)^{\frac{1}{4}}$

$$y^2 = (x+9)^{\frac{1}{4} \times 2} = (x+9)^{\frac{1}{2}}$$

$$y^4 = x+9$$

$$4y^3 dy = dx$$

$$\int \frac{y}{1+y^2} \cdot 4y^3 dy$$

$$4 \int \frac{y^4}{1+y^2} dy$$

$$\begin{array}{r} y^2 - 1 \\ 1+y^2 \overline{) y^4} \end{array}$$

$$\ominus y^4 \oplus y^2$$

$$-y^2$$

$$\oplus -y^2 \oplus 1$$

$$1$$

$$4 \int \left[y^2 - 1 + \frac{1}{1+y^2} \right] dy$$

$$4 \left[\frac{y^3}{3} - y + \tan^{-1} y \right] + c$$

$$4 \left[\frac{((x+9)^{\frac{1}{4}})^3}{3} - (x+9)^{\frac{1}{4}} + \tan^{-1} (x+9)^{\frac{1}{4}} \right] + c$$

$$\int \frac{dx}{x\sqrt{1+\sqrt{x}}}$$

Let $y = \sqrt{1+\sqrt{x}}$

$$y^2 = 1 + \sqrt{x}$$

$$y^2 - 1 = \sqrt{x}$$

$$x = (y^2 - 1)^2$$

$$dx = 2(y^2 - 1) 2y dy$$

$$\int \frac{2(y^2 - 1) 2y dy}{(y^2 - 1)^2 y}$$

$$\int \frac{4}{y^2 - 1} dy$$

$$= 4 \frac{1}{2(1)} \ln \left| \frac{y-1}{y+1} \right| + C$$

$$= 2 \ln \left| \frac{\sqrt{1+\sqrt{x}} - 1}{\sqrt{1+\sqrt{x}} + 1} \right| + C$$

$$\int \frac{1}{x \left(x^{\frac{1}{6}} - 1 \right)^{\frac{1}{2}}} dx$$

Let $y = \left(x^{\frac{1}{6}} - 1 \right)^{\frac{1}{2}}$

$$y^2 = x^{\frac{1}{6}} - 1$$

$$y^2 + 1 = x^{\frac{1}{6}}$$

$$x = (y^2 + 1)^6$$

$$dx = 6(y^2 + 1)^5 \cdot 2y \, dy$$

$$\int \frac{1}{(y^2 + 1)^6} \cdot 6(y^2 + 1)^5 \cdot 2y \, dy$$

$$\int \frac{12y}{y^2 + 1} \, dy$$

$$= 12 \tan^{-1} y + C$$

$$= 12 \tan^{-1} \left(x^{\frac{1}{6}} - 1 \right)^{\frac{1}{2}} + C$$

$$\int \frac{dx}{\sqrt{1 + \sqrt[3]{x} \sqrt[3]{x}}}$$

جذر داخل جذر

Let $y = \sqrt{1 + \sqrt[3]{x}}$

$$y^2 = 1 + \sqrt[3]{x}$$

$$\sqrt[3]{x} = y^2 - 1$$

$$x = (y^2 - 1)^3$$

$$dx = 3(y^2 - 1)^2 \cdot 2y \, dy$$

$$\int \frac{3(y^2 - 1)^2 \cdot 2y}{y \cdot (y^2 - 1)} \, dy$$

$$6 \int (y^2 - 1) \, dy$$

$$= 6 \left[\frac{y^3}{3} - y \right] + C$$

$$= 6 \left[\frac{(\sqrt{1 + \sqrt[3]{x}})^3}{3} - \sqrt{1 + \sqrt[3]{x}} \right] + C$$

$$\int \frac{x + 6\sqrt{x} + 9}{x^2\sqrt{x} + 4x\sqrt{x}} dx$$

Let $y = \sqrt{x}$

$y^2 = x$

$2y dy = dx$

$\longrightarrow (y^2)^2 = x^2$
 $y^4 = x^2$

$$\int \frac{y^2 + 6y + 9}{y^4 y + 4 y^2 y} \cdot 2y dy$$

$$2 \int \frac{y^2 + 6y + 9}{y^5 + 4y^3} y dy$$

$$2 \int \frac{y^2 + 6y + 9}{y^3 (y^2 + 4)} y dy$$

$$2 \int \frac{y^2 + 6y + 9}{y^2 (y^2 + 4)} dy$$

$$\frac{y^2 + 6y + 9}{y^2 (y^2 + 4)} = \frac{A}{y} + \frac{B}{y^2} + \frac{Cy + D}{y^2 + 4}$$

* $y^2 (y^2 + 4)$

$$y^2 + 6y + 9 = Ay(y^2 + 4) + B(y^2 + 4) + (Cy + D)y^2$$

$y=0$

$$0 + 0 + 9 = 0 + B(4) + 0 \rightarrow B = \frac{9}{4}$$

Equate the coeff. of y^3

$$0 = A + C$$

$$C = -A \rightarrow \textcircled{1}$$

سكالة

Equate the coeff. of y^2

$$1 = B + D$$

$$1 = \frac{9}{4} + D$$

$$1 - \frac{9}{4} = D \rightarrow D = -\frac{5}{4}$$

سكالة

Equate the coeff. of y

$$6 = 4A \rightarrow A = \frac{6}{4}$$

$$A = \frac{3}{2}$$

in $\textcircled{1}$

$$C = -\frac{3}{2}$$

$$\int \left(\frac{\frac{3}{2}}{y} + \frac{\frac{9}{4}}{y^2} + \frac{\frac{-3}{2}y - \frac{5}{4}}{y^2 + 4} \right) dy$$

$\frac{3}{2} \ln|y| + \frac{9}{4} \frac{y^{-1}}{-1} - 3 \frac{1}{2} \ln(y^2 + 4) - \frac{5}{4} \frac{1}{2} \tan^{-1}\left(\frac{y}{2}\right) + C$

$\sqrt{x}$

$$\int \frac{2x}{1 + \sqrt[3]{3x^2 - 1}} dx$$

Let $y = \sqrt[3]{3x^2 - 1}$

$$y^3 = 3x^2 - 1$$

$$3y^2 dy = 6x dx$$

$$\frac{3}{6} y^2 dy = x dx \Rightarrow \frac{1}{2} y^2 dy = x dx$$

$$\int \frac{2}{1+y} \cdot \frac{1}{2} y^2 dy$$

$$\int \frac{y^2}{1+y} dy$$

$$\int \left[y-1 + \frac{1}{1+y} \right] dy$$

$$\frac{y^2}{2} - y + \ln|1+y| + c$$

$$\frac{(\sqrt[3]{3x^2-1})^2}{2} - \sqrt[3]{3x^2-1} + \ln|1+\sqrt[3]{3x^2-1}| + c$$



Text book question



$$\int \frac{x^3}{\sqrt[3]{x^2 + 1}} dx$$

Let $y = \sqrt[3]{x^2 + 1}$

$$y^3 = x^2 + 1 \longrightarrow x^2 = y^3 - 1$$

$$3y^2 dy = 2x dx$$

$$\frac{3}{2} y^2 dy = x dx$$

$$\int \frac{x^2 \cdot \boxed{x} \cdot \boxed{dx}}{\sqrt[3]{x^2 + 1}}$$

$$\int \frac{y^3 - 1}{y} \cdot \frac{3}{2} y^2 dy$$

$$\frac{3}{2} \int (y^4 - y) dy$$

$$\frac{3}{2} \left[\frac{y^5}{5} - \frac{y^2}{2} \right] + C$$

$$\frac{3}{2} \left[\frac{(\sqrt[3]{x^2 + 1})^5}{5} - \frac{(\sqrt[3]{x^2 + 1})^2}{2} \right] + C$$

$$\int \frac{t \sqrt{1+2t^2}}{1+t^2} dt$$

Let $y = \sqrt{1+2t^2}$

$$y^2 = 1+2t^2$$

$$\longrightarrow y^2 - 1 = 2t^2$$

$$2y dy = 4t dt$$

$$t^2 = \frac{1}{2}(y^2 - 1)$$

$$= \frac{1}{2}y^2 - \frac{1}{2}$$

$$\frac{2}{4} y dy = t dt$$

$$\int \frac{y}{1 + \frac{1}{2}y^2 - \frac{1}{2}} \cdot \frac{2}{4} y dy$$

$$\frac{\frac{1}{2} + \frac{1}{2}y^2}{\frac{1}{2}(1+y^2)}$$

$$\int \frac{y}{\frac{1}{2}(1+y^2)} \cdot \frac{1}{2} y dy$$

$$\int \frac{y^2}{1+y^2} dy$$

$$\int \left[1 + \frac{-1}{1+y^2} \right] dy$$

$$y - \tan^{-1} y + C$$

$$\sqrt{1+2t^2} - \tan^{-1} \sqrt{1+2t^2} + C$$

$$\begin{array}{r} 1+y^2 \overline{) y^2} \\ \underline{-y^2} \\ 1 \end{array}$$

$$\int \ln(1 + \sqrt{x}) dx$$

حل بدرس سابق

$$t = \sqrt{x}$$

$$dt = \frac{1}{2\sqrt{x}} dx$$

$$2dt = \frac{1}{\sqrt{x}} dx$$

$$\int \frac{\sqrt{x}}{\sqrt{x}} \ln(1 + \sqrt{x}) dx$$

$$2 \int t \ln(1+t) dt$$

$$u = \ln(1+t) \quad dv = t dt$$

$$du = \frac{1}{1+t} dt \quad v = \frac{t^2}{2}$$

$$\frac{1}{2} \ln(1+t) t^2 - \frac{1}{2} \int \frac{t^2}{1+t} dt$$

$$\int \left[t-1 + \frac{1}{1+t} \right] dt$$

$$\frac{t^2}{2} - t + \ln|1+t| + C$$

$$\Rightarrow 2 \left[\frac{1}{2} \ln(1+t) t^2 - \frac{1}{2} \left(\frac{t^2}{2} - t + \ln|1+t| \right) \right] + C$$

حل درس اليوم

$$\text{Let } y = \sqrt{x}$$

$$y^2 = x$$

$$2y dy = dx$$

$$2 \int y \ln(1+y) dy$$

$$u = \ln(1+y) \quad dv = y dy$$

$$du = \frac{1}{1+y} dy \quad v = \frac{y^2}{2}$$

$$\frac{1}{2} \ln(1+y) y^2 - \frac{1}{2} \int \frac{y^2}{1+y} dy$$

$$\frac{y-1}{1+y} \sqrt{\frac{y^2}{y^2+y}}$$

$$\int \left[y-1 + \frac{1}{1+y} \right] dy$$

$$\frac{y^2}{2} - y + \ln|1+y| + C$$

$$2 \left[\frac{1}{2} \ln(1+y) y^2 - \frac{1}{2} \left(\frac{y^2}{2} - y + \ln|1+y| \right) \right] + C$$

$$\int \tan^{-1} \sqrt{x} \, dx$$

Let $y = \sqrt{x}$
 $y^2 = x$
 $2y \, dy = dx$

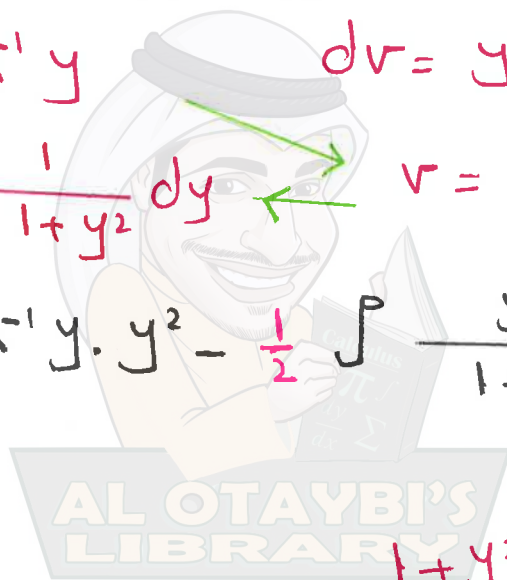
$$\int \tan^{-1} y \cdot 2y \, dy$$

$$2 \int y \tan^{-1} y \, dy$$

$u = \tan^{-1} y$ $dv = y \, dy$

$du = \frac{1}{1+y^2} dy$ $v = \frac{y^2}{2}$

$$\frac{1}{2} \tan^{-1} y \cdot y^2 - \frac{1}{2} \int \frac{y^2}{1+y^2} dy$$



$$\frac{1}{1+y^2} = \frac{y^2}{y^2+1} - \frac{1}{y^2+1}$$

$$\int \left[1 + \frac{-1}{1+y^2} \right] dy$$

$$y - \tan^{-1} y + c$$

$$\Rightarrow 2 \left[\frac{1}{2} \tan^{-1} y \cdot y^2 - \frac{1}{2} (y - \tan^{-1} y) \right] + c$$

$$2 \left[\frac{1}{2} \tan^{-1} \sqrt{x} (\sqrt{x})^2 - \frac{1}{2} (\sqrt{x} - \tan^{-1} \sqrt{x}) \right] + c$$

$$\int \frac{\sin(2x)}{\sqrt{5 - \cos x}} dx$$

$$\int \frac{2 \sin x \cos x}{\sqrt{5 - \cos x}} dx$$

$$y = \sqrt{5 - \cos x}$$

$$y^2 = 5 - \cos x \longrightarrow \cos x = 5 - y^2$$

$$2y dy = \sin x dx$$

$$\int \frac{2(5 - y^2)}{y} 2y dy$$

$$4 \int (5 - y^2) dy$$

$$4 \left[5y - \frac{y^3}{3} \right] + C$$

$$4 \left[5\sqrt{5 - \cos x} - \frac{(\sqrt{5 - \cos x})^3}{3} \right] + C$$

$$\int \ln(x + \sqrt{x}) dx$$

$$y = \sqrt{x}$$

$$y^2 = x$$

$$2y dy = dx$$

$$\int \ln(y^2 + y) 2y dy$$

$$2 \int y \ln(y^2 + y) dy$$

$$u = \ln(y^2 + y) \quad dv = y dy$$

$$du = \frac{2y+1}{y^2+y} dy \quad v = \frac{y^2}{2}$$

$$\frac{1}{2} \ln(y^2 + y) y^2 - \frac{1}{2} \int y^2 \frac{2y+1}{y^2+y} dy$$

$$\int \frac{2y^3 + y^2}{y^2 + y} dy$$

$$\int \frac{y^2(2y+1)}{y(y+1)} dy$$

$$\int \frac{2y^2 + y}{y+1} dy$$

$$\begin{array}{r} 1+y \overline{) 2y^2 + y} \\ \underline{2y^2 + y} \\ 0 \end{array}$$

$$\int [2y - 1 + \frac{1}{y+1}] dy$$

$$\frac{2y^2}{2} - y + \ln|y+1| + c$$

$$\frac{1}{2} \ln(y^2 + y) y^2 - \frac{1}{2} [y^2 - y + \ln|y+1|] + c$$

$\sqrt{x}$

$$\int \frac{x}{x+1+\sqrt[3]{x+1}} dx$$

Let $y = \sqrt[3]{x+1}$

$$y^3 = x+1$$

$$y^3 - 1 = x$$

$$3y^2 dy = dx$$

$$\int \frac{y^3 - 1}{y^3 + y} \cdot 3y^2 dy$$

$$3 \int \frac{y^3 - 1}{y(y^2 + 1)} y^2 dy$$

$$3 \int \frac{y^4 - y}{y^2 + 1} dy$$

$$3 \int \left[y^2 - 1 + \frac{-y+1}{y^2+1} \right] dy$$

$$3 \int \left[y^2 - 1 + \frac{\frac{1}{2}(2y) + 1}{y^2 + 1} \right] dy$$

$$\frac{\frac{1}{2}(2y)}{y^2 + 1} + \frac{1}{y^2 + 1}$$

$$3 \left[\frac{y^3}{3} - y - \frac{1}{2} \ln|y^2 + 1| + \tan^{-1} y \right] + C$$

$$\sqrt[3]{x+1}$$

$$\begin{array}{r} y^2 - 1 \\ y^4 - y \\ \hline y^4 + y^2 \\ \hline -y^2 - y \\ \hline -y^2 + 1 \\ \hline y + 1 \end{array}$$



Tricky Question !!



$$\int \frac{x^{\frac{3}{2}}}{x+1} dx$$

Let $y = x^{\frac{1}{2}}$

$$y^2 = x$$

$$2y dy = dx$$

$$y^3 = x^{\frac{3}{2}}$$

$$\int \frac{y^3}{y^2+1} 2y dy$$

$$2 \int \frac{y^4}{y^2+1} dy$$

$$2 \int \left[y^2 - 1 + \frac{1}{y^2+1} \right] dy$$

$$2 \left[\frac{y^3}{3} - y + \tan^{-1} y \right] + c$$

$$2 \left[\frac{(x^{\frac{1}{2}})^3}{3} - x^{\frac{1}{2}} + \tan^{-1} x^{\frac{1}{2}} \right] + c$$

$$\begin{array}{r} y^2 - 1 \\ 1 + y^2 \overline{) y^4} \\ \underline{\ominus y^4 \oplus y^2} \\ -y^2 \\ \underline{\oplus -y^2 \oplus 1} \\ 1 \end{array}$$



Improper integral



هو تكامل محدد يحتوي على نقاط عدم اتصال (نقاط حرجية)

خطوات الحل

(1) حل التكامل بدون أطراف.

$\infty, -\infty$

(2) أدور الحرج

أصفار المقام بشرط انتمائها
لفترة حدود التكامل

شروط أساسية !!

(1) حرج واحد فقط.

(2) يكون على الطرف.

(3) أشيل الحرج وأحط مكانه t ثم اخذ اللمت له

الناتج

$\infty, -\infty, DNE$

رقم

Divergent

Convergent

Determine whether the following integral is convergent or divergent , if it possible evaluate it

$$\int_{-\infty}^1 \frac{dx}{x^2 + 2x + 5} \longrightarrow \text{ما كثر أضعاف للمقام}$$

$$\begin{aligned} I_1 &= \int \frac{dx}{x^2 + 2x + 5} \\ &= \int \frac{dx}{(x+1)^2 + 4} \quad \begin{array}{l} \xrightarrow{\quad} x^2 + 2x + 5 \\ (x+1)^2 - 1 + 5 \\ (x+1)^2 + 4 \end{array} \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) + C \end{aligned}$$

$$\lim_{t \rightarrow -\infty} \int_t^1 \frac{dx}{x^2 + 2x + 5}$$

$$\lim_{t \rightarrow -\infty} \left[\frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) \right]_t^1$$

$$\frac{1}{2} \lim_{t \rightarrow -\infty} \left[\tan^{-1} \left(\frac{2}{2} \right) - \tan^{-1} \left(\frac{t+1}{2} \right) \right]$$

$$\begin{aligned} &\frac{1}{2} \left[\frac{\pi}{4} - \tan^{-1}(-\infty) \right] = \frac{1}{2} \left[\frac{\pi}{4} - \left(-\frac{\pi}{2} \right) \right] \\ &= \frac{1}{2} \left[\frac{\pi}{4} + \frac{2\pi}{4} \right] = \frac{1}{2} \frac{3\pi}{4} = \frac{3\pi}{8} \end{aligned}$$

$$\Rightarrow \therefore \int_{-\infty}^1 \frac{dx}{x^2 + 2x + 5} \text{ is } \underline{\underline{\text{convergent}}}$$

Determine whether the following integral is convergent or divergent, if it possible evaluate it

$$\int_{-\infty}^{-4} \frac{x+1}{x^2+2x} dx$$

$$I_1 = \int \frac{\frac{1}{2} (2x+2)}{x^2+2x} dx = \frac{1}{2} \ln|x^2+2x| + c$$

$$\lim_{t \rightarrow -\infty} \int_t^{-4} \frac{x+1}{x^2+2x} dx$$

$$\lim_{t \rightarrow -\infty} \left[\frac{1}{2} \ln|x^2+2x| \right]_t^{-4}$$

$$\frac{1}{2} \lim_{t \rightarrow -\infty} \left[\ln|16-8| - \ln|t^2+2t| \right]$$

$$\frac{1}{2} \lim_{t \rightarrow -\infty} \left[\ln 8 - \ln|t(t+2)| \right]$$

$$\frac{1}{2} \left[\ln 8 - \ln|-\infty(-\infty)| \right]$$

$$\frac{1}{2} [\ln 8 - \ln \infty] = -\infty$$

$$\Rightarrow \therefore \int_{-\infty}^{-4} \frac{x+1}{x^2+2x} dx \text{ is } \underline{\underline{\text{divergent}}}$$

Determine whether the following integral is convergent or divergent , if it possible evaluate it

$$\int_1^{\infty} \frac{dx}{x\sqrt{6x-1}}$$

∞
 $x\sqrt{6x-1} = 0$
 $x=0 \quad x=\frac{1}{6}$
 $\notin [1, \infty) \quad \notin [1, \infty)$

$$I_1 = \int \frac{dx}{x\sqrt{6x-1}}$$

$$y = \sqrt{6x-1}$$

$$y^2 = 6x-1$$

$$y^2 + 1 = 6x$$

$$x = \frac{1}{6}(y^2 + 1)$$

$$dx = \frac{1}{6} 2y dy$$

$$\int \frac{\cancel{\frac{1}{6}} 2y}{\cancel{\frac{1}{6}}(y^2+1) \cancel{y}} dy = \int \frac{2}{y^2+1} dy$$

$$= 2 \tan^{-1} y + C = 2 \tan^{-1} \sqrt{6x-1} + C$$

$$\lim_{t \rightarrow \infty} \int_1^t \frac{dx}{x\sqrt{6x-1}}$$

$$\lim_{t \rightarrow \infty} [2 \tan^{-1} \sqrt{6x-1}]_1^t = 2 \lim_{t \rightarrow \infty} [\tan^{-1} \sqrt{6t-1} - \tan^{-1} \sqrt{6-1}]$$

$$= 2 \left[\frac{\pi}{2} - \tan^{-1} \sqrt{5} \right]$$

$$\Rightarrow \therefore \int_1^{\infty} \frac{dx}{x\sqrt{6x-1}} \text{ is } \underline{\text{convergent}}$$



Text book question



Determine whether the following integral is convergent or divergent, if it converges evaluate it.

$$\int_1^{\infty} \frac{e^{-\frac{1}{x}}}{x^2} dx$$

$$I_1 = \int \frac{e^{-\frac{1}{x}}}{x^2} dx$$

$$\text{Let } t = \frac{1}{x}$$

$$dt = -\frac{1}{x^2} dx \rightarrow -dt = \frac{1}{x^2} dx$$

$$-\int e^{-t} dt = -\frac{e^{-t}}{-1} + C = e^{-\frac{1}{x}} + C$$

$$\lim_{t \rightarrow \infty} \int_1^t \frac{e^{-\frac{1}{x}}}{x^2} dx$$

$$\lim_{t \rightarrow \infty} \left[e^{-\frac{1}{x}} \right]_1^t$$

$$\begin{aligned} \lim_{t \rightarrow \infty} [e^{-\frac{1}{t}} - e^{-1}] &= e^{-\frac{1}{\infty}} - e^{-1} \\ &= e^0 - e^{-1} \\ &= 1 - e^{-1} \end{aligned}$$

$$\therefore \int_1^{\infty} \frac{e^{-\frac{1}{x}}}{x^2} dx \text{ is converges to } 1 - e^{-1}$$

Determine whether the following integral is convergent or divergent, if it possible evaluate it

$$\int_0^1 \frac{dx}{e^x - 1}$$

$$\begin{aligned} e^x - 1 &= 0 \\ e^x &= 1 \\ x &= 0 \in [0, 1] \end{aligned}$$

$$\begin{aligned} I_1 &= \int \frac{dx}{e^x - 1} * \frac{e^{-x}}{e^{-x}} \\ &= \int \frac{e^{-x}}{1 - e^{-x}} dx = \ln|1 - e^{-x}| + C \end{aligned}$$

$$\lim_{t \rightarrow 0^+} \int_t^1 \frac{dx}{e^x - 1}$$

$$\lim_{t \rightarrow 0^+} \left[\ln|1 - e^{-x}| \right]_t^1$$

$$\lim_{t \rightarrow 0^+} \left[\ln|1 - e^{-1}| - \ln|1 - e^{-t}| \right]$$

$$= \ln|1 - e^{-1}| - \ln|1 - e^0|$$

$$= \ln|1 - e^{-1}| - \ln|1 - 1|$$

$$= \ln|1 - e^{-1}| - \ln 0$$

$$= \ln|1 - e^{-1}| - (-\infty) = \infty$$

$$\Rightarrow \therefore \int_0^1 \frac{dx}{e^x - 1} \text{ is } \underline{\underline{\text{divergent}}}$$

Determine whether the following integral is convergent or divergent , if it possible evaluate it

$$\int_0^{\infty} \frac{dx}{e^{2x} + 9}$$

$$I_1 = \int \frac{dx}{e^{2x} + 9} \quad * \quad \frac{e^{-2x}}{e^{-2x}}$$

$$= \int \frac{e^{-2x}}{1 + 9e^{-2x}} dx$$

$$= \frac{-1}{18} \int \frac{-18 e^{-2x}}{1 + 9e^{-2x}} dx = \frac{-1}{18} \ln|1 + 9e^{-2x}| + C$$

$$\lim_{t \rightarrow \infty} \int_0^t \frac{dx}{e^{2x} + 9}$$

$$\lim_{t \rightarrow \infty} \left[\frac{-1}{18} \ln|1 + 9e^{-2x}| \right]_0^t$$

$$= \frac{-1}{18} \lim_{t \rightarrow \infty} \left[\ln|1 + 9e^{-2t}| - \ln|1 + 9e^0| \right]$$

$$= \frac{-1}{18} \left[\ln|1 + 9e^{-\infty}| - \ln|1 + 9| \right]$$

$$= \frac{-1}{18} \left[\ln|1| - \ln 10 \right] = \frac{1}{18} \ln 10$$

$$\Rightarrow \therefore \int_0^{\infty} \frac{dx}{e^{2x} + 9} \text{ is convergent}$$

Determine whether the following integral is convergent or divergent, and if convergent, evaluate it

$$\int_1^3 \frac{dx}{x^2 \sqrt{x^2 - 1}}$$

$0 \notin [1, 3]$
 $-1 \notin [1, 3]$
 $1 \in [1, 3]$

$$I_1 = \int \frac{dx}{x^2 \sqrt{x^2 - 1}}$$

Let $x = \sec \theta$
 $dx = \sec \theta \tan \theta d\theta$



$$= \int \frac{\sec \theta \tan \theta}{\sec^2 \theta \sqrt{\sec^2 \theta - 1}} d\theta$$

$$\sqrt{\tan^2 \theta} = \tan \theta$$

$$= \int \frac{\cancel{\sec \theta} \cancel{\tan \theta}}{\sec^2 \theta \cancel{\tan \theta}} d\theta = \int \frac{1}{\sec \theta} d\theta$$

$$= \int \cos \theta d\theta = \sin \theta + C = \frac{\sqrt{x^2 - 1}}{x} + C$$

$$\lim_{t \rightarrow 1^+} \int_t^3 \frac{dx}{x^2 \sqrt{x^2 - 1}}$$

$$\lim_{t \rightarrow 1^+} \left[\frac{\sqrt{x^2 - 1}}{x} \right]_t^3 = \lim_{t \rightarrow 1^+} \left[\frac{\sqrt{8}}{3} - \frac{\sqrt{t^2 - 1}}{t} \right]$$

$$= \frac{\sqrt{8}}{3} - \frac{0}{1} = \frac{\sqrt{8}}{3}$$

$$\Rightarrow \therefore \int_1^3 \frac{dx}{x^2 \sqrt{x^2 - 1}} \text{ is Convergent}$$

Determine whether the following integral is convergent or divergent, and if possible evaluate it

$$\int_1^4 \frac{dx}{\sqrt{8-x^2+2x}} \rightarrow \begin{aligned} & -[x^2-2x-8] \\ & -[(x+2)(x-4)] \\ & x=-2 \quad x=4 \end{aligned}$$

$\in [1, 4]$

$$I_1 = \int \frac{dx}{\sqrt{8-x^2+2x}}$$

$$= \int \frac{dx}{\sqrt{9-(x-1)^2}}$$

قانون

$$= \sin^{-1}\left(\frac{x-1}{3}\right) + C$$

$$\begin{aligned} & -[x^2-2x-8] \\ & -[(x-1)^2-1-8] \\ & -[(x-1)^2-9] \\ & 9-(x-1)^2 \end{aligned}$$

$$\lim_{t \rightarrow 4^-} \int_1^t \frac{dx}{\sqrt{8-x^2+2x}}$$

$$\lim_{t \rightarrow 4^-} \left[\sin^{-1}\left(\frac{x-1}{3}\right) \right]_1^t$$

$$\lim_{t \rightarrow 4^-} \left[\sin^{-1}\left(\frac{t-1}{3}\right) - \sin^{-1} 0 \right]$$

$$= \sin^{-1}\left(\frac{3}{3}\right) - 0$$

$$= \sin^{-1} 1 = \frac{\pi}{2}$$

$$\Rightarrow \therefore \int_1^4 \frac{dx}{\sqrt{8-x^2+2x}} \text{ is convergent}$$

Determine whether the following integral is convergent or divergent, if it possible evaluate it

$$\int_0^1 \frac{1}{x + \sqrt{x}} dx \longrightarrow 0 \in [0,1]$$

$$I_1 = \int \frac{1}{x + \sqrt{x}} dx$$

Let $y = \sqrt{x}$
 $y^2 = x$
 $2y dy = dx$

$$\int \frac{1}{y^2 + y} \cdot 2y dy = \int \frac{1}{y(y+1)} 2y dy$$
$$= 2 \ln|y+1| + C = 2 \ln|\sqrt{x} + 1| + C$$

$$\lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{x + \sqrt{x}} dx$$

$$\lim_{t \rightarrow 0^+} \left[2 \ln|\sqrt{x} + 1| \right]_t^1 = 2 \lim_{t \rightarrow 0^+} \left[\ln|2| - \ln|t+1| \right]$$

$$= 2 \left[\ln 2 - \ln 1 \right] = 2 \ln 2$$

$$\Rightarrow \therefore \int_0^1 \frac{1}{x + \sqrt{x}} dx \text{ is convergent}$$

Determine whether the following integral is convergent or divergent , if it possible evaluate it

$$\int_{\ln 3}^{\infty} \frac{e^x}{e^{2x} - 3e^x + 2} dx \longrightarrow (\infty)$$

$$I_1 = \int \frac{e^x}{(e^x)^2 - 3e^x + 2} dx$$

Let $u = e^x$
 $du = e^x dx$

$$= \int \frac{1}{u^2 - 3u + 2} du$$

$$\frac{1}{(u-1)(u-2)} = \frac{A}{u-1} + \frac{B}{u-2}$$

* $(u-1)(u-2)$

$$1 = A(u-2) + B(u-1)$$

$u=1$

$$1 = A(-1) + 0 \longrightarrow A = -1$$

$u=2$

$$1 = 0 + B(1) \longrightarrow B = 1$$

$$\int \left[\frac{-1}{u-1} + \frac{1}{u-2} \right] du$$

$$= -\ln|u-1| + \ln|u-2| + C$$

$$= -\ln|e^x - 1| + \ln|e^x - 2| + C$$

← See next page plz!

$$\lim_{t \rightarrow \infty} \int_{\ln 3}^t \frac{e^x}{e^{2x} - 3e^x + 2} dx$$

$$\lim_{t \rightarrow \infty} \left[-\ln|e^x - 1| + \ln|e^x - 2| \right]_{\ln 3}^t$$

$$\lim_{t \rightarrow \infty} \left[\ln \left(\frac{e^x - 2}{e^x - 1} \right) \right]_{\ln 3}^t$$

$$\lim_{t \rightarrow \infty} \left[\ln \left(\frac{e^t - 2}{e^t - 1} \right) - \ln \left(\frac{e^{\ln 3} - 2}{e^{\ln 3} - 1} \right) \right]$$

استثناء e
 $\nwarrow$ $\nearrow$
 $\ln$ $\ln$

$$\begin{aligned} &= \ln(1) - \ln\left(\frac{3-2}{3-1}\right) \\ &= 0 - \ln\left(\frac{1}{2}\right) \\ &= -\ln\left(\frac{1}{2}\right) \end{aligned}$$

$\Rightarrow \therefore \int_{\ln 3}^{\infty} \frac{e^x}{e^{2x} - 3e^x + 2} dx$ is convergent

إذا عتب أكثر
 من $\ln$ حلج
 $\downarrow$
 أعلينهم
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 ودره

Determine whether the following integral is convergent or divergent , and if it possible evaluate it

$$\int_0^1 \frac{\ln x}{\sqrt{x}} dx$$

$$I_1 = \int \frac{\ln x}{\sqrt{x}} dx$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$dv = \frac{1}{\sqrt{x}} dx = x^{-\frac{1}{2}}$$

$$v = \frac{x^{\frac{1}{2}}}{\frac{1}{2}} = 2x^{\frac{1}{2}}$$

$$2 \ln x \cdot x^{\frac{1}{2}} - 2 \int x^{\frac{1}{2}} \frac{1}{x} dx \rightarrow x^{\frac{1}{2}-1} = x^{-\frac{1}{2}}$$

$$2 \ln x \cdot x^{\frac{1}{2}} - 2 \int x^{-\frac{1}{2}} dx$$

$$2 \ln x \cdot x^{\frac{1}{2}} - 2 \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$2 \ln x \cdot x^{\frac{1}{2}} - 4x^{\frac{1}{2}} + C$$

$$\lim_{t \rightarrow 0^+} \int_t^1 \frac{\ln x}{\sqrt{x}} dx$$

$$\lim_{t \rightarrow 0^+} \left[2 \ln x \cdot x^{\frac{1}{2}} - 4x^{\frac{1}{2}} \right]_t^1$$



$$\lim_{t \rightarrow 0^+} \left[(0 - 4) - (2 \ln t \cdot t^{\frac{1}{2}} - 4t^{\frac{1}{2}}) \right]$$

$$\lim_{t \rightarrow 0^+} \ln t \cdot t^{\frac{1}{2}} \quad (-\infty) 0$$

$$\lim_{t \rightarrow 0^+} \frac{\ln t}{\frac{1}{t^{\frac{1}{2}}}} \quad \frac{-\infty}{\frac{1}{0}} \quad \frac{-\infty}{\infty}$$

$$\lim_{t \rightarrow 0^+} \frac{\ln t}{t^{-\frac{1}{2}}}$$

L'Hop

$$\lim_{t \rightarrow 0^+} \frac{\frac{1}{t}}{\frac{-1}{2} t^{-\frac{3}{2}}} \quad \swarrow \searrow$$

$$\lim_{t \rightarrow 0^+} \frac{2t^{\frac{3}{2}}}{-t}$$

$$\lim_{t \rightarrow 0^+} \frac{2t^{\frac{1}{2}}}{-1} = 0$$

$$\Rightarrow \therefore \lim_{t \rightarrow 0^+} \left[-4 - (2 \ln t \cdot t^{\frac{1}{2}} - 4t^{\frac{1}{2}}) \right] = -4 - (0 - 0) = -4$$

$$\Rightarrow \therefore \int \frac{\ln x}{\sqrt{x}} dx \text{ is } \underline{\underline{\text{convergent}}}$$

Determine whether the following integral is convergent or divergent, if it possible evaluate it

$$\int_{-\infty}^{\infty} e^{-x} dx$$

$\nearrow -\infty$
 $\searrow \infty$

$$I_1 = \int e^{-x} dx = \frac{e^{-x}}{-1} + c = -e^{-x} + c$$

$$\int_{-\infty}^0 e^{-x} dx + \int_0^{\infty} e^{-x} dx$$

$$\lim_{t \rightarrow -\infty} \int_t^0 e^{-x} dx + \lim_{t \rightarrow \infty} \int_0^t e^{-x} dx$$

$$\lim_{t \rightarrow -\infty} [-e^{-x}]_t^0 + \lim_{t \rightarrow \infty} [-e^{-x}]_0^t$$

$$-\lim_{t \rightarrow -\infty} [e^0 - e^{-t}] - \lim_{t \rightarrow \infty} [e^{-t} - e^0]$$

$$- [1 - e^{\infty}] - [e^{-\infty} - 1]$$

$$- [1 - \infty] - [0 - 1]$$

$$= \infty$$

$$\Rightarrow \therefore \int_{-\infty}^{\infty} e^{-x} dx \text{ is divergent}$$



Remark from Calculus A



في مسائل اللمت , عندما يكون ناتج التعويض

$$\frac{\text{عدد}}{0} = \infty$$

يتم دراسة اشارة الانفتي

$$\lim_{x \rightarrow 5^+} \frac{3}{5-x} = \frac{3}{0^-} = -\infty$$

$$\lim_{x \rightarrow 5^-} \frac{3}{5-x} = \frac{3}{0^+} = +\infty$$

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Determine whether the following integral is convergent or divergent, if it possible evaluate it

$$\int_2^{\infty} \frac{x}{(x^2-4)^2} dx$$

$\nearrow \infty$
 $\searrow 2 \in [2, \infty)$
 $\searrow -2 \notin [2, \infty)$

$$I_1 = \int \frac{x}{(x^2-4)^2} dx = \int \underline{x} (x^2-4)^{-2} \underline{dx}$$

Let $u = x^2 - 4$

$du = 2x dx \longrightarrow \frac{1}{2} du = x dx$

$$\frac{1}{2} \int u^{-2} du = \frac{1}{2} \underline{u^{-1}} + c = \underline{-\frac{1}{2} \frac{1}{u} + c}$$

$$= -\frac{1}{2} \frac{1}{x^2-4} + c$$

$$\int_2^3 \frac{x}{(x^2-4)^2} dx + \int_3^{\infty} \frac{x}{(x^2-4)^2} dx$$

$\lim_{t \rightarrow 2^+} \int_t^3 \frac{x}{(x^2-4)^2} dx + \lim_{t \rightarrow \infty} \int_3^t \frac{x}{(x^2-4)^2} dx$

$\lim_{t \rightarrow 2^+} \left[-\frac{1}{2} \frac{1}{x^2-4} \right]_t^3 + \lim_{t \rightarrow \infty} \left[-\frac{1}{2} \frac{1}{x^2-4} \right]_3^t$

$\lim_{t \rightarrow 2^+} \left[\frac{1}{9-4} - \frac{1}{t^2-4} \right] - \frac{1}{2} \lim_{t \rightarrow \infty} \left[\frac{1}{t^2-4} - \frac{1}{9-4} \right]$

$\frac{1}{\infty} = 0$

$$= -\frac{1}{2} \left[\frac{1}{5} - \infty \right] - \frac{1}{2} \left[0 - \frac{1}{5} \right]$$

$$= \infty$$

$\Rightarrow \int_2^{\infty} \frac{x}{(x^2-4)^2} dx$ is divergent

Determine whether the following integral is convergent or divergent , if it possible evaluate it

$$\int_0^4 (2-x)^{-\frac{4}{3}} dx \quad \longrightarrow 2 \in [0,4]$$

$$I_1 = \int (2-x)^{-\frac{4}{3}} dx$$

$$= \frac{(2-x)^{-\frac{1}{3}}}{(-1) \cdot \frac{-1}{3}} + C = \frac{3}{(2-x)^{\frac{1}{3}}} + C$$

$$\int_0^2 (2-x)^{-\frac{4}{3}} dx + \int_2^4 (2-x)^{-\frac{4}{3}} dx$$

$$\lim_{t \rightarrow 2^-} \int_0^t (2-x)^{-\frac{4}{3}} dx + \lim_{t \rightarrow 2^+} \int_t^4 (2-x)^{-\frac{4}{3}} dx$$

$$\lim_{t \rightarrow 2^-} \left[\frac{3}{(2-x)^{\frac{1}{3}}} \right]_0^t + \lim_{t \rightarrow 2^+} \left[\frac{3}{(2-x)^{\frac{1}{3}}} \right]_t^4$$

$$3 \lim_{t \rightarrow 2^-} \left[\frac{1}{(2-t)^{\frac{1}{3}}} - \frac{1}{(2)^{\frac{1}{3}}} \right] + 3 \lim_{t \rightarrow 2^+} \left[\frac{1}{(-2)^{\frac{1}{3}}} - \frac{1}{(2-t)^{\frac{1}{3}}} \right]$$

$\frac{1}{0^+} \rightarrow \infty$
 $\frac{1}{0^-} \rightarrow -\infty$

$$= \infty + \infty = \infty$$

$$\Rightarrow \therefore \int_0^4 (2-x)^{-\frac{4}{3}} dx \text{ is divergent}$$

Determine whether the following integral is convergent or divergent, if it possible evaluate it

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin(2x)}{\sqrt[3]{\sin^4 x}} dx \longrightarrow 0 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$I_1 = \int \frac{\sin(2x)}{(\sin x)^{\frac{4}{3}}} dx = \int \frac{2 \sin x \cos x}{(\sin x)^{\frac{4}{3}}} dx$$

$$= 2 \int (\sin x)^{1 - \frac{4}{3}} \cos x dx$$

$$= 2 \int (\sin x)^{-\frac{1}{3}} \cos x dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$= 2 \int u^{-\frac{1}{3}} du = 2 \cdot \frac{u^{\frac{2}{3}}}{\frac{2}{3}} + C$$

$$= 3 (\sin x)^{\frac{2}{3}} + C$$

$$\int_{-\frac{\pi}{2}}^0 \frac{\sin(2x)}{\sqrt[3]{\sin^4 x}} dx + \int_0^{\frac{\pi}{2}} \frac{\sin(2x)}{\sqrt[3]{\sin^4 x}} dx$$

$$\lim_{t \rightarrow 0^-} \int_{-\frac{\pi}{2}}^t \frac{\sin(2x)}{\sqrt[3]{\sin^4 x}} dx + \lim_{t \rightarrow 0^+} \int_t^{\frac{\pi}{2}} \frac{\sin(2x)}{\sqrt[3]{\sin^4 x}} dx$$

$$\lim_{t \rightarrow 0^-} \left[3 (\sin x)^{\frac{2}{3}} \right]_{-\frac{\pi}{2}}^t + \lim_{t \rightarrow 0^+} \left[3 (\sin x)^{\frac{2}{3}} \right]_t^{\frac{\pi}{2}}$$

$$3 \lim_{t \rightarrow 0^-} \left[(\sin t)^{\frac{2}{3}} - (\sin(-\frac{\pi}{2}))^{\frac{2}{3}} \right] + 3 \lim_{t \rightarrow 0^+} \left[(\sin \frac{\pi}{2})^{\frac{2}{3}} - (\sin t)^{\frac{2}{3}} \right]$$

$$3 \left[0 - (-1)^{\frac{2}{3}} \right] + 3 \left[1^{\frac{2}{3}} - 0 \right]$$

$$= -3 + 3 = 0 \implies \therefore \text{Convergent}$$

Determine whether the following integral is convergent or divergent, if it possible evaluate it

$$\int_0^{\infty} \frac{1}{(x-1)^2} dx$$

∞
 $1 \in [0, \infty)$

$$I_1 = \int \frac{1}{(x-1)^2} dx = \int (x-1)^{-2} dx = \frac{(x-1)^{-1}}{-1} + C = \frac{-1}{x-1} + C$$

$$\begin{aligned} & \int_0^1 \frac{1}{(x-1)^2} dx + \int_1^2 \frac{1}{(x-1)^2} dx + \int_2^{\infty} \frac{1}{(x-1)^2} dx \\ &= \lim_{t \rightarrow 1^-} \int_0^t \frac{1}{(x-1)^2} dx + \lim_{t \rightarrow 1^+} \int_t^2 \frac{1}{(x-1)^2} dx + \lim_{t \rightarrow \infty} \int_2^t \frac{1}{(x-1)^2} dx \\ &= \lim_{t \rightarrow 1^-} \left[\frac{-1}{x-1} \right]_0^t + \lim_{t \rightarrow 1^+} \left[\frac{-1}{x-1} \right]_t^2 + \lim_{t \rightarrow \infty} \left[\frac{-1}{x-1} \right]_2^t \\ &= \lim_{t \rightarrow 1^-} \left[\frac{1}{t-1} - \frac{1}{-1} \right] - \lim_{t \rightarrow 1^+} \left[\frac{1}{1} - \frac{1}{t-1} \right] - \lim_{t \rightarrow \infty} \left[\frac{1}{1} - \frac{1}{t-1} \right] \end{aligned}$$

$$= \infty - (-\infty) = \infty + \infty = \infty$$

$$\Rightarrow \int_0^{\infty} \frac{1}{(x-1)^2} dx \text{ is } \underline{\underline{\text{divergent}}}$$

لعبة !!



$$\int_{-9}^9 \frac{x}{x^2 - 1} dx$$

$$\begin{aligned} x^2 - 1 &= 0 \\ (x-1)(x+1) &= 0 \\ x &= 1 \quad x = -1 \\ &\in [-9, 9] \quad \in [-9, 9] \end{aligned}$$

$$\int_{-9}^{-1} \frac{x}{x^2 - 1} dx + \int_{-1}^0 \frac{x}{x^2 - 1} dx + \int_0^1 \frac{x}{x^2 - 1} dx + \int_1^9 \frac{x}{x^2 - 1} dx$$

$$\int_{-\infty}^{\infty} \frac{x}{x^2 - 1} dx$$

$$\begin{aligned} x^2 - 1 &= 0 \\ (x-1)(x+1) &= 0 \\ x &= 1 \quad x = -1 \\ &\in (-\infty, \infty) \quad \in (-\infty, \infty) \end{aligned}$$

$$\int_{-\infty}^{-2} \frac{x}{x^2 - 1} dx + \int_{-2}^{-1} \frac{x}{x^2 - 1} dx + \int_{-1}^0 \frac{x}{x^2 - 1} dx + \int_0^1 \frac{x}{x^2 - 1} dx + \int_1^2 \frac{x}{x^2 - 1} dx + \int_2^{\infty} \frac{x}{x^2 - 1} dx$$



Text book question



Find the values of p for which the integral converges and evaluate the integral for those values of p .

$$\int_e^{\infty} \frac{1}{x(\ln x)^p} dx$$

$$I_1 = \int \frac{1}{x(\ln x)^p} dx$$

$$\text{Let } u = \ln x \\ du = \frac{1}{x} dx$$

$$\int \frac{1}{u^p} du = \int u^{-p} du = \frac{u^{-p+1}}{-p+1} + C$$

$$= \frac{1}{-p+1} (\ln x)^{-p+1} + C$$

$$\lim_{t \rightarrow \infty} \int_e^t \frac{1}{x(\ln x)^p} dx$$

$$\lim_{t \rightarrow \infty} \left[\frac{1}{-p+1} (\ln x)^{-p+1} \right]_e^t$$

$$\frac{1}{-p+1} \lim_{t \rightarrow \infty} [(\ln t)^{-p+1} - 1] = \frac{1}{-p+1} [0 - 1]$$

$$\text{when } -p+1 < 0 \longrightarrow 1 < p$$

$$\therefore \int_e^{\infty} \frac{1}{x(\ln x)^p} dx \text{ is converges to } \frac{-1}{-p+1} \text{ when } 1 < p$$

$$\int_e^{\infty} \frac{dx}{x (\ln x)^3}$$

Nov 2016

قراءة

$$I_1 = \int \frac{dx}{x (\ln x)^3}$$

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\int \frac{1}{u^3} du = \int u^{-3} du = \frac{u^{-2}}{-2} + C$$

$$= -\frac{1}{2} \frac{1}{u^2} + C = -\frac{1}{2} \frac{1}{(\ln x)^2} + C$$

$$\lim_{t \rightarrow \infty} \int_e^t \frac{dx}{x (\ln x)^3}$$

$$\lim_{t \rightarrow \infty} \left[-\frac{1}{2} \frac{1}{(\ln x)^2} \right]_e^t$$

$$-\frac{1}{2} \lim_{t \rightarrow \infty} \left[\frac{1}{(\ln t)^2} - \frac{1}{(\ln e)^2} \right]$$

$$-\frac{1}{2} \left[\frac{1}{\infty} - \frac{1}{1} \right]$$

$$-\frac{1}{2} [0 - 1] = \frac{1}{2}$$

$\Rightarrow \therefore \int_e^{\infty} \frac{dx}{x (\ln x)^3}$ is convergent.



Multiple choice



$$\int_{-\infty}^3 e^{\frac{x}{3}} dx$$

Oct 2017

- a) converges to e^3 . b) converges to $9e$.
c) converges to $e/3$. d) diverges.
e) converges to $3e$.

$$I_1 = \int e^{\frac{x}{3}} dx = \frac{e^{\frac{x}{3}}}{\frac{1}{3}} + C = 3e^{\frac{x}{3}} + C$$

$$\lim_{t \rightarrow -\infty} \int_t^3 e^{\frac{x}{3}} dx$$

$$\lim_{t \rightarrow -\infty} \left[3e^{\frac{x}{3}} \right]_t^3$$

$$3 \lim_{t \rightarrow -\infty} \left[e^1 - e^{\frac{t}{3}} \right]$$

$$3 \left[e - \underbrace{e^{-\infty}}_0 \right] = 3e \quad (\underline{e})$$



Multiple choice



$$\int_1^2 \frac{1}{\sqrt{(x-1)^5}} dx$$

Oct 2017

a) diverges.

b) converges to $-2/3$.

c) converges to $\sqrt{3}$.

d) converges $4/3$.

e) converges to $2/3$.

$$\begin{aligned} I_1 &= \int (x-1)^{-\frac{5}{2}} dx = \frac{(x-1)^{-\frac{3}{2}}}{-\frac{3}{2}} + C \\ &= -\frac{2}{3} \frac{1}{(x-1)^{\frac{3}{2}}} + C \end{aligned}$$

$$\lim_{t \rightarrow 1^+} \int_t^2 \frac{1}{\sqrt{(x-1)^5}} dx$$

$$\lim_{t \rightarrow 1^+} \left[-\frac{2}{3} \frac{1}{(x-1)^{\frac{3}{2}}} \right]_t^2$$

$$-\frac{2}{3} \lim_{t \rightarrow 1^+} \left[\frac{1}{1} - \frac{1}{(t-1)^{\frac{3}{2}}} \right]$$

$$-\frac{2}{3} \left[1 - \frac{1}{0^+} \right]$$

$$-\frac{2}{3} [1 - \infty] = \infty \rightarrow \therefore \text{diverges (a) ✓}$$



Multiple choice



Which one of the following integrals is not an improper integral ?

a) $\int_0^3 \frac{1}{2-x} dx$

b) $\int_0^{\pi/2} \tan^{-1} x dx.$

Oct 2017

c) $\int_0^{\pi/2} \sec x dx.$

d) $\int_0^{\infty} \frac{x^2 - 1}{x^2 + 4} dx.$

e) $\int_0^1 \ln(x^2) dx.$

(b) ✓





Multiple choice



$\int \frac{1}{x + \sqrt[3]{x}} dx$ is equal to

a) $\frac{3}{2} \ln |\sqrt[3]{x} + 1| + C.$

b) $3 \ln (\sqrt[3]{x^2} + 1) + C.$

c) $\frac{3}{2} \tan^{-1} (\sqrt[3]{x^2} + 1) + C.$

d) $-\frac{3}{2} \tan^{-1} |\sqrt[3]{x^2} + 1| + C.$

e) $\frac{3}{2} \ln (\sqrt[3]{x^2} + 1) + C.$

Oct 2017

$$\int \frac{1}{x + \sqrt[3]{x}} dx$$

Let

$$y = \sqrt[3]{x}$$

$$y^3 = x$$

$$3y^2 dy = dx$$

$$\int \frac{1}{y^3 + y} 3y^2 dy$$

$$\int \frac{\cancel{3y^2}}{\cancel{y} (y^2 + 1)} dy$$

$$3 \left(\frac{1}{2} \right) \int \frac{y}{y^2 + 1} dy = \frac{3}{2} \ln |y^2 + 1| + C$$
$$= \frac{3}{2} \ln |(\sqrt[3]{x})^2 + 1| + C$$

(e) ✓



Calculus B

Quiz # 6



$$\int (e^{3x} + 2^x - e^{\frac{x}{2}} + 3^{6x}) dx$$

$$= \frac{e^{3x}}{3} + \frac{2^x}{(1) \ln 2} - \frac{e^{\frac{x}{2}}}{\frac{1}{2}} + \frac{3^{6x}}{6 \ln 3} + C$$

$$\int \sec^2 x e^{\tan x} dx$$

Let $t = \tan x$
 $dt = \sec^2 x dx$

$$\int e^t dt = e^t + C = e^{\tan x} + C$$

$$\int \frac{\cos x}{\sin^2 x + 4} dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$\int \frac{1}{u^2 + 4} du = \frac{1}{2} \tan^{-1}\left(\frac{u}{2}\right) + C = \frac{1}{2} \tan^{-1}\left(\frac{\sin x}{2}\right) + C$$

$$\int \frac{e^{\frac{1}{x}}}{x^3} dx$$

Let $t = \frac{1}{x}$

$$dt = -\frac{1}{x^2} dx \rightarrow -dt = \frac{1}{x^2} dx$$

$$\int \frac{e^{\frac{1}{x}}}{x \cdot x^2} dx$$

$$\int t e^t dt$$

$u = t$ $dv = e^t dt$
 $du = dt$ $v = e^t$

$$t e^t - \int e^t dt$$

$$t e^t - e^t + C \rightarrow \frac{1}{x} e^{\frac{1}{x}} - e^{\frac{1}{x}} + C$$

